

Jermaine Shirlee Jr.

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EDUCATION

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Science, Pursuing Minor in Mathematics

Champaign, IL

Aug. 2026 – Current

Current Coursework: Probability & Statistics for Computer Science, Linear Algebra w/ Computational Application, Computer Architecture, Software Design Lab

College of DuPage

Grainger Engineering Pathways, GPA: 3.76/4.0

Glen Ellyn, IL

Aug. 2024 – July 2026

Relevant Coursework: Advanced C++ w/ Data Structures, Discrete Mathematics, Linear Algebra, Introduction to Java, Programming Logic & Technique, Introduction to Linux

EXPERIENCE

Machine Learning Research Intern

Fermi National Accelerator Laboratory

June 2026 – Aug. 2026

Batavia, IL

- Trained a ResNet18 convolutional neural network to predict muon beam initial momentum from 2D spatial distribution histograms, achieving a mean absolute error under 1 MeV/c
- Developed a C++ data macro to process G4Beamline particle simulation output and generate 100 statistically varied training samples per momentum value using random event sampling
- Simulated muon particle beams across a range of initial momentum values using G4Beamline (Geant4), applying a 99% transmission threshold to ensure data standardization across all ten virtual detectors

Summer Exchange Program Intern

Fermi National Accelerator Laboratory

June 2025 – Aug. 2025

Batavia, IL

- Selected for competitive DOE-funded cohort, completed 6-week summer program at Fermilab and Brookhaven National Laboratory, 3 weeks each
- Actively learned about particle physics, climate modeling, AI/Machine Learning, and quantum computing through lectures, simulations, and lab-based projects
- Collaborated with peers and lab scientists to analyze research challenges and projects, presented findings

Caretaker

Angel's Homecare Services

Jan. 2025 – Aug. 2026

Lombard, IL

- Caretaker of grandmother diagnosed with severe Alzheimer's Disease during school year
- Assist client with ADL (Activities of Daily Living)

HONORS

C++ Proficiency Certificate | College of DuPage

Issued May 2026

- Completed College of DuPage's C++ programming curriculum, covering object-oriented programming, data structures, and algorithms; awarded language proficiency certificate

Academic Honors | College of DuPage

Issued Fall 2024, Fall 2025, Spring 2026

- Awarded each semester to students with a 3.50 to 4.00 GPA who take at least six credit hours

S-STEM Scholarship | National Science Foundation

Issued Aug. 2024

- Selected for prestigious full-ride scholarship for high achieving STEM students
- Nominated by College of DuPage and selected among thousands of students nation-wide to attend the AAAS S-STEM REC conference

TECHNICAL SKILLS

Languages: Java, Python, C++

Frameworks: PyTorch, ROOT, OpenCV, G4Beamline/Geant4

Developer Tools: Git, VS Code, Visual Studio, Eclipse

Libraries: pandas, NumPy, Matplotlib